

Program 8: Build a front-end web application using React and apply the React Render method.

Aim: To Build a front-end web application using React and apply the React Render method.

Algorithm:

Step 1: Install Node.js and Vite (If Not Installed).

- a. Check if **Node.js** and **npm** are installed.

```
node -v
```

```
npm -v
```

[Note: If not installed, download from [Node.js official site](https://nodejs.org/en/) and install it.]

Step 2: Create a React Project Using Vite.

- a. Open **VS Code** and launch a terminal.
- b. Run the following command to create a new React project using **Vite**:

```
npm create vite@latest my-app --template react
```

- c. **Navigate to the project folder** and install dependencies:

```
cd my-app
```

```
npm install
```

step 3: Modify main.jsx to Implement React Render Method

- a. Open **VS Code** and navigate to src/main.jsx.
- b. Replace the existing code with the following:

src/main.jsx (Manual React Render)

```
import React from "react";
```

```
import ReactDOM from "react-dom/client";
```

```
let count = 0; // Initialize count
```

```
// Function to update count and re-render the app
```

```
const updateCount = () => {
```

```
  count += 1; // Increment count
```

```
  renderApp(); // Re-render manually
```

```
};
```

```
// Function to render the React app
```

```

const renderApp = () => {
  ReactDOM.createRoot(document.getElementById("root")).render(
    <React.StrictMode>
      <div style={{ textAlign: "center", marginTop: "50px" }}>
        <h1>🚀 Welcome to Vite + React</h1>
        <p>This is a simple front-end React application using React Render.</p>
        <h2>Count: {count}</h2> { /* Display count */ }
        <button
          onClick={updateCount}
          style={{ padding: "10px 20px", fontSize: "16px", cursor: "pointer" }}
        >
          Click to Increase Count
        </button>
      </div>
    </React.StrictMode>
  );
};

// Initial Render
renderApp();

```

step 4: Ensure index.html Has a Root Element

Check public/index.html to ensure there is a div with id="root":

public/index.html

```

<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

```

```
<title>Vite + React</title>

</head>

<body>

  <div id="root"></div> <!-- React will render here -->

  <script type="module" src="/src/main.jsx"></script>

</body>

</html>
```

Step 5: Run the React Application

- a. Start the development server:

```
npm run dev
```

- b. Open **http://localhost:5173/** in your browser.
 - c. Click the **"Click to Increase Count"** button.
 - d. The count will update because **React re-renders the entire UI manually**.
-

Output:



Welcome to Vite + React

This is a simple front-end React application using React Render.

Count: 5

Click to Increase Count